

An Age-Structured SEIRS-V Epidemic Model with Seasonal Forcing, Prevalence-Dependent Behavioral Adaptation, and Prior Predictive Ensemble Uncertainty Quantification

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GitHub: <https://github.com/TanmaySonawane/SEIRS-V>

Live Dashboard: <https://huggingface.co/spaces/T22S/seirs-v-model>

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Abstract

Classical compartmental models of infectious disease dynamics assume homogeneous population mixing, constant transmission rates, permanent post-infection immunity, and precisely known parameters. This report presents a novel age-structured SEIRS-V model that addresses all four limitations simultaneously. The model partitions a population of $N = 700,000$ individuals into two age groups connected by a 2×2 empirical contact matrix. Transmission varies seasonally via a cosine forcing function, and a prevalence-dependent behavioral dampening factor $\Phi(t) = 1/(1 + \kappa I_{\text{total}}/N)$ reduces contact rates as the infectious population grows. This behavioral adaptation term is a novel contribution not present in any of the reviewed literature. Nine uncertain biological parameters are drawn from calibrated prior distributions using Latin Hypercube Sampling, producing a 500-member prior predictive ensemble with 95% credible interval bands. The basic reproduction number R_0 is derived analytically via the next-generation matrix. Under baseline parameters, $R_0 \approx 3.23$, and the behavioral parameter κ is shown analytically to have no effect on R_0 . Four structured validation checks confirm model correctness. The model is deployed as an interactive dashboard and REST API at the links above.

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1. Introduction

Mathematical models of infectious disease transmission have served as essential tools for public health decision-making since the foundational work of Kermack and McKendrick in 1927 [1]. The canonical Susceptible-Infectious-Recovered (SIR) model reduces epidemic dynamics to three coupled ordinary differential equations parameterized by a transmission rate β and recovery rate γ . Despite its elegance, the SIR framework rests on assumptions that consistently fail in real epidemic settings: populations do not mix homogeneously across age groups, immunity is rarely permanent, transmission rates fluctuate with season, and parameter values are never known with precision.

This project develops a systematic extension of the SIR framework motivated by five papers spanning the foundational, intermediate, and advanced compartmental modeling literature, as well as by visual experimentation with NetLogo agent-based simulations. The resulting model, designated SEIRS-V, incorporates age-structured transmission via an empirically motivated contact matrix, seasonal forcing, a novel prevalence-dependent behavioral adaptation term, waning natural and vaccine-derived immunity, and a prior predictive ensemble that replaces point-estimate parameters with probability distributions.

The contributions of this work are threefold. First, the behavioral dampening factor $\Phi(t) = 1/(1 + \kappa I_{\text{total}}/N)$ is a novel formulation not present in any of the reviewed papers. Second, the model combines age structure, seasonality, behavioral adaptation, and vaccine waning simultaneously. Third, the full pipeline is deployed as a public REST API and interactive Streamlit dashboard accessible via the links on the title page.

The report is organized as follows. Section 2 reviews the literature. Section 3 discusses NetLogo simulations. Section 4 derives the model specification. Section 5 derives R_0 via the next-generation matrix. Section 6 describes probabilistic parameterization and Latin Hypercube Sampling. Section 7 covers numerical methods. Section 8 presents validation. Section 9 presents results. Sections 10 and 11 discuss implications and limitations.

2. Literature Review and Foundational Papers

2.1. The Basic SIR Framework

Two foundational papers: Tolles and Luong [2] and an introduction to deterministic compartmental models [3], establish the canonical SIR model:

$$\frac{dS}{dt} = -\beta SI \tag{1}$$

$$\frac{dI}{dt} = \beta SI - \gamma I \tag{2}$$

$$\frac{dR}{dt} = \gamma I \tag{3}$$

where β is the transmission rate and $\gamma = 1/T_{\text{inf}}$ is the recovery rate. The term βSI represents mass-action incidence: every susceptible contacts every infectious individual with equal probability. The basic reproduction number is $R_0 = \beta/\gamma$: when $R_0 > 1$ the epidemic grows; when $R_0 < 1$ it dies out.

The key limitations identified in these papers are: closed population, perfectly homogeneous mixing, permanent post-infection immunity, and exactly known parameters. Each limitation is addressed by a specific extension in this project.

2.2. Bridge Papers and Mathematical Extensions

2.2.1. Vital Dynamics and the SEIR Framework

Blackwood and Childs [4] introduce vital dynamics and an Exposed compartment E , transforming SIR into SEIR. The inclusion of births Λ and deaths μ produces endemic equilibria: the disease can persist indefinitely rather than always burning out. The Exposed compartment introduces a latency period $1/\sigma$ between infection and becoming infectious, critical for diseases where transmission begins before symptom onset. This paper directly motivated the Exposed compartment and vital dynamics in this model.

2.2.2. Waning Immunity, Non-Linear Incidence, and Age Structure

Hethcote [5] provides mathematical foundations for three extensions. First, the SIRS extension with waning immunity at rate ω :

$$\frac{dR}{dt} = \gamma I - \omega R - \mu R \quad (4)$$

This enables recurrent epidemic waves. Second, non-linear saturating incidence $\beta SI/(1 + \alpha I)$ models behavioral saturation at high prevalence. This concept, though formulated differently here, motivated the behavioral dampening term in this project. Third, age-structured contact matrices β_{ij} are introduced, directly motivating the two-group contact matrix used here.

2.2.3. Multi-Compartment Models and the Next-Generation Matrix

Brauer [6] provides the theoretical framework for computing R_0 in complex multi-compartment systems via the next-generation matrix (NGM). The key result, formalized by van den Driessche and Watmough [7], is:

$$R_0 = \rho(\mathbf{FV}^{-1}) \quad (5)$$

where $\mathbf{F}$ contains new infection rates, $\mathbf{V}$ contains transition rates, and $\rho(\cdot)$ is the spectral radius. This framework is applied in full in Section 5.

3. NetLogo Simulations as Visual Precursors

Prior to developing the mathematical model, two NetLogo agent-based simulations were studied. The *Basic epiDEM* model simulates individual agents moving on a grid, changing color with disease state. This illustrated the role of initial seed size, the stochasticity of individual transmission, and the visual emergence of spatial clustering. The *Travel and Control* model extended this to multiple spatial patches connected by travel, illustrating how mobility drives inter-regional spread and how travel restrictions reshape epidemic curves.

Both simulations revealed their own limitations: no analytical R_0 , no parameter sweeps, no age structure, no uncertainty quantification. They built intuition for what the ODE terms

represent: the $S \rightarrow E \rightarrow I \rightarrow R$ flow corresponds directly to watching agents change color; the contact matrix corresponds to agent encounter probabilities, but could not provide the mathematical rigor that the ODE model supplies.

4. Model Specification

4.1. Population and Compartments

The model partitions $N = 700,000$ individuals into two age groups: youth (ages 0 - 17, $N_1 = 200,000$) and adults (ages 18+, $N_2 = 500,000$). Each individual belongs to exactly one of five compartments per group: Susceptible S_i , Exposed E_i , Infectious I_i , Recovered R_i , Vaccinated V_i . The complete state vector is:

$$\mathbf{y}(t) = [S_1, E_1, I_1, R_1, V_1, S_2, E_2, I_2, R_2, V_2]^\top \in \mathbb{R}^{10} \quad (6)$$

The demographic assumption $\Lambda_i = \mu N_i$ ensures $S_i + E_i + I_i + R_i + V_i = N_i$ for all $t \geq 0$ (verified in Section 8).

4.2. Seasonal Forcing

Respiratory pathogens exhibit well-documented seasonal transmission variation [8]. The baseline transmission rate β_0 is modulated by a cosine function with a 365-day period:

$$\beta(t) = \beta_0 \left(1 + \varepsilon \cos\left(\frac{2\pi t}{365}\right) \right) \quad (7)$$

where $\varepsilon \in [0, 1]$ is the seasonality amplitude. At $t = 0$ (winter), $\beta(t) = \beta_0(1 + \varepsilon)$; at $t = 182$ days (summer), $\beta(t) = \beta_0(1 - \varepsilon)$. Setting $\varepsilon = 0$ recovers constant transmission.

4.3. Prevalence-Dependent Behavioral Adaptation

Classical compartmental models assume constant contact rates regardless of epidemic severity. COVID-19 demonstrated that populations substantially reduce contact in response to rising case counts independently of government mandates [9]. None of the five reviewed papers implement this mechanism. The following formulation is introduced as a novel contribution:

$$\Phi(t) = \frac{1}{1 + \kappa I_{\text{total}}(t)/N} \quad (8)$$

where $I_{\text{total}}(t) = I_1(t) + I_2(t)$ and $\kappa \geq 0$ is the behavioral response strength. $\Phi(t)$ is bounded in $(0, 1]$, equals 1 when no disease is present, and is strictly decreasing in prevalence. Setting $\kappa = 0$ recovers the standard model without adaptation. Table 1 illustrates contact reduction at selected prevalence levels.

Table 1: Behavioral dampening factor $\Phi(t)$ at selected prevalence levels under $\kappa = 5$.

Prevalence I/N	$\Phi(t)$	Contact reduction
0%	1.000	0%
1%	0.952	4.8%
5%	0.800	20%
10%	0.667	33%
20%	0.500	50%

4.4. Contact Matrix and Force of Infection

The force of infection on group i incorporates all three innovations:

$$\lambda_i(t) = \beta(t) \cdot \Phi(t) \cdot \sum_{j=1}^2 c_{ij} \frac{I_j(t)}{N_j} \quad (9)$$

The contact rates c_{ij} are drawn from the POLYMOD social contact survey [10]. POLYMOD (Improving Public Health Policy in Europe through Modelling and Economic Evaluation of Interventions) is a landmark multi-country survey recording all social contacts over one day stratified by age. The resulting contact matrices are the standard reference for parameterizing age-structured epidemic models. For our two-group aggregation:

$$\mathbf{C} = \begin{pmatrix} 9.0 & 3.0 \\ 1.5 & 4.5 \end{pmatrix} \quad (\text{contacts per person per day}) \quad (10)$$

Diagonal entries represent within-group contacts dominated by school (youth) and workplace (adult) mixing. Off-diagonal entries represent cross-group contacts. The reciprocity constraint $c_{ij}N_i = c_{ji}N_j$ is approximately satisfied; $c_{12}N_1 = 600,000 \approx c_{21}N_2 = 750,000$, with the small asymmetry within POLYMOD uncertainty bounds.

In matrix form with $\mathbf{f}(t) = [I_1/N_1, I_2/N_2]^\top$:

$$[\lambda_1(t), \lambda_2(t)]^\top = \beta(t) \cdot \Phi(t) \cdot \mathbf{C} \mathbf{f}(t) \quad (11)$$

4.5. The Complete ODE System

The full SEIRS-V model consists of ten coupled ODEs. For $i \in \{1, 2\}$:

$$\frac{dS_i}{dt} = \mu N_i + \omega R_i + \omega_v V_i - \lambda_i S_i - \nu_i S_i - \mu S_i \quad (12)$$

$$\frac{dE_i}{dt} = \lambda_i S_i + (1 - \delta) \lambda_i V_i - \sigma E_i - \mu E_i \quad (13)$$

$$\frac{dI_i}{dt} = \sigma E_i - \gamma I_i - \mu I_i \quad (14)$$

$$\frac{dR_i}{dt} = \gamma I_i - \omega R_i - \mu R_i \quad (15)$$

$$\frac{dV_i}{dt} = \nu_i S_i - (1 - \delta) \lambda_i V_i - \omega_v V_i - \mu V_i \quad (16)$$

The biological interpretation of each term follows. In equation (12): μN_i is births; ωR_i is natural immunity waning; $\omega_v V_i$ is vaccine waning; $-\lambda_i S_i$ is new infections; $-\nu_i S_i$ is vaccination outflow; $-\mu S_i$ is natural death. In equation (13): $\lambda_i S_i$ is newly infected susceptibles entering latency; $(1 - \delta)\lambda_i V_i$ is breakthrough infections among vaccinated individuals reduced by efficacy δ ; $-\sigma E_i$ is progression to infectious. In equation (14): σE_i arrives from exposed; $-\gamma I_i$ is recovery. In equation (15): $-\omega R_i$ is immunity waning enabling reinfection. In equation (16): $\nu_i S_i$ is vaccination inflow; $-(1 - \delta)\lambda_i V_i$ is breakthrough outflow to E_i .

5. The Basic Reproduction Number via the Next-Generation Matrix

5.1. Motivation and Disease-Free Equilibrium

For a single-group model without vaccination, $R_0 = \beta/\gamma$ is immediately readable from the equations. In a multi-group model with vaccination, this scalar formula fails: transmission operates across groups at different rates and the susceptible pool is partitioned between unvaccinated and partially protected vaccinated individuals. The next-generation matrix (NGM) framework of van den Driessche and Watmough [7] provides a general solution.

The NGM is evaluated at the disease-free equilibrium (DFE): the steady state with $E_i = I_i = 0$ for all i . Setting $dS_i/dt = dV_i/dt = 0$ at the DFE yields:

$$S_i^* = \frac{\mu N_i(\omega_v + \mu)}{\mu(\omega_v + \mu) + \nu_i \mu}, \quad V_i^* = \frac{\nu_i S_i^*}{\omega_v + \mu} \quad (17)$$

One verifies $S_i^* + V_i^* = N_i$, confirming demographic consistency.

5.2. The New Infection Matrix $\mathbf{F}$

Let the infected compartments be ordered $[E_1, E_2, I_1, I_2]$. The matrix $\mathbf{F}$ contains only terms creating new infections, flows into E compartments from contact with infectious individuals. At the DFE, $I_{\text{total}} = 0$ so $\Phi = 1$ and $\beta(t) = \beta_0$:

$$\mathbf{F} = \begin{pmatrix} 0 & 0 & f_{13} & f_{14} \\ 0 & 0 & f_{23} & f_{24} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (18)$$

Bottom rows are zero because new infections enter E , not I , compartments. Left columns are zero because exposed individuals do not generate new infections.

As a worked example, f_{13} new exposed youth per infectious youth is:

$$f_{13} = \beta_0 \cdot c_{11} \cdot \frac{S_1^* + (1 - \delta)V_1^*}{N_1} \quad (19)$$

This is: transmission probability per contact $\times$ within-group contacts per day $\times$ fraction of youth effectively susceptible at the DFE. The effective susceptibles $S_1^* + (1 - \delta)V_1^*$ account for the fact that vaccinated individuals face $(1 - \delta)$ of the normal infection probability. The remaining entries follow by substitution:

$$f_{ij+2} = \beta_0 \cdot c_{ij} \cdot \frac{S_i^* + (1 - \delta)V_i^*}{N_j}, \quad i, j \in \{1, 2\} \quad (20)$$

5.3. The Transition Matrix $\mathbf{V}$

$\mathbf{V}$ contains all other flows out of infected compartments: progression, recovery, and death. Positive diagonal entries represent leaving; negative off-diagonal entries represent arriving from another infected compartment:

$$\mathbf{V} = \begin{pmatrix} \sigma + \mu & 0 & 0 & 0 \\ 0 & \sigma + \mu & 0 & 0 \\ -\sigma & 0 & \gamma + \mu & 0 \\ 0 & -\sigma & 0 & \gamma + \mu \end{pmatrix} \quad (21)$$

Since $\mathbf{V}$ is block lower-triangular, its inverse is analytic:

$$\mathbf{V}^{-1} = \begin{pmatrix} \frac{1}{\sigma + \mu} & 0 & 0 & 0 \\ 0 & \frac{1}{\sigma + \mu} & 0 & 0 \\ \frac{\sigma}{(\sigma + \mu)(\gamma + \mu)} & 0 & \frac{1}{\gamma + \mu} & 0 \\ 0 & \frac{\sigma}{(\sigma + \mu)(\gamma + \mu)} & 0 & \frac{1}{\gamma + \mu} \end{pmatrix} \quad (22)$$

Each entry has a probabilistic interpretation. $1/(\sigma + \mu) \approx 1/\sigma = 5$ days is the average time in the exposed compartment. $1/(\gamma + \mu) \approx 1/\gamma = 10$ days is the average infectious duration. The off-diagonal entry $\sigma/[(\sigma + \mu)(\gamma + \mu)]$ is the probability of surviving latency times average infectious duration.

5.4. The Next-Generation Matrix and R_0

The NGM $\mathbf{K} = \mathbf{F}\mathbf{V}^{-1}$ collapses to a 2×2 matrix whose (i, j) entry is the total number of group- i infections caused by one infectious person in group j :

$$K_{ij} = \beta_0 \cdot c_{ij} \cdot \frac{\sigma}{(\sigma + \mu)(\gamma + \mu)} \cdot \frac{S_i^* + (1 - \delta)V_i^*}{N_j} \quad (23)$$

The factor $\sigma/[(\sigma + \mu)(\gamma + \mu)]$ is (probability of surviving latency) $\times$ (average infectious duration) ≈ 9.977 days under default parameters. Numerical values under default parameters:

$$\mathbf{K} \approx \begin{pmatrix} 2.955 & 0.394 \\ 1.230 & 1.476 \end{pmatrix} \quad (24)$$

One infectious youth infects 2.955 other youth and 1.230 adults; one infectious adult infects 0.394 youth and 1.476 adults.

Proposition 1. *The basic reproduction number is the dominant eigenvalue of $\mathbf{K}$:*

$$R_0 = \frac{K_{11} + K_{22}}{2} + \sqrt{\left(\frac{K_{11} - K_{22}}{2}\right)^2 + K_{12}K_{21}} \quad (25)$$

Step-by-step computation under default parameters:

$$\begin{aligned} (K_{11} + K_{22})/2 &= 2.216, & K_{12}K_{21} &= 0.394 \times 1.230 = 0.485 \\ R_0 &= 2.216 + \sqrt{0.547 + 0.485} = 2.216 + 1.016 \approx \mathbf{3.23} \end{aligned}$$

Proposition 2 (Behavioral adaptation does not affect R_0). *The parameter κ does not affect R_0 .*

Proof. At the DFE, $I_{\text{total}} = 0$ by definition, so $\Phi = 1/(1 + \kappa \cdot 0/N) = 1$ for any $\kappa \geq 0$. Since $\Phi = 1$, it factors out of every entry of $\mathbf{F}$ and cancels. Therefore $\mathbf{K}$ is independent of κ , and so is $R_0 = \rho(\mathbf{K})$. $\square$

This result is verified numerically by computing R_0 at $\kappa \in \{0, 1, 5, 10, 50, 100\}$; all values agree to ten decimal places.

6. Probabilistic Parameterization and Latin Hypercube Sampling

6.1. From Point Estimates to Prior Predictive Ensemble

Treating biological parameters as fixed constants implies a precision that field data cannot support. This project adopts a prior predictive ensemble: each uncertain parameter is assigned a prior distribution, and 500 complete parameter sets are drawn and each used to integrate the ODE system once. The spread of the resulting 500 trajectories represents forecast uncertainty under the prior.

This differs from a stochastic model (where transmission events themselves are random, requiring different numerical methods such as Gillespie or Euler-Maruyama) and from a Bayesian posterior ensemble (which would require observed case data and MCMC). The prior predictive approach is appropriate in the absence of real data and is standard practice in published epidemic modeling during early outbreak phases [9].

6.2. Prior Distributions

Nine parameters are treated as uncertain. Table 2 summarizes their distributions.

Table 2: Prior distributions for the nine uncertain parameters.

Parameter	Distribution	Parameters	Rationale
β_0	Trunc. Normal	(0.035, 0.005), > 0	Symmetric measurement error
ε	Uniform	(0.10, 0.30)	Literature range
κ	Uniform	(2.0, 10.0)	Deep behavioral uncertainty
σ	Gamma	(25, 0.008)	Positive, right-skewed
γ	Gamma	(25, 0.004)	Positive, right-skewed
ω	Uniform	(1/270, 1/90)	Immunity 3 - 9 months
δ	Beta	(32, 8)	Bounded in $[0, 1]$, mean = 0.80
ν_1	Uniform	(0.001, 0.004)	Policy uncertainty
ν_2	Uniform	(0.0005, 0.002)	Policy uncertainty

Truncated Normal is used for β_0 (symmetric uncertainty, must remain positive). Gamma is used for biological rates σ and γ (strictly positive, right-skewed clinical distributions [11]). Beta is used for vaccine efficacy δ (bounded in $[0, 1]$ by definition). Uniform is used for

parameters where the range is plausible but no value within it is clearly preferred, the maximum entropy distribution on a bounded interval.

Post-sampling, two validity filters are applied. First, hard biological bounds are enforced for each parameter. Second, the ratio filter:

$$0.3 \leq \frac{\sigma}{\gamma} \leq 4.0 \quad (26)$$

This prevents biologically implausible combinations such as 1-day incubation paired with 25-day infectious period. It is a pragmatic substitute for a Gaussian copula correlation structure between σ and γ , noted as a limitation in Section 11.

6.3. Latin Hypercube Sampling

Drawing 500 parameter sets from nine distributions via simple random sampling risks clustering in high-probability regions. Latin Hypercube Sampling (LHS), introduced by McKay, Beckman, and Conover [12], resolves this through stratification. Unlike simple random sampling which can cluster near the distribution mean, LHS guarantees that every equal-probability region of the prior is represented.

Algorithm 1 Latin Hypercube Sampling for Prior Predictive Ensemble

- 1: Divide $[0, 1]$ into n equal intervals of width $1/n$
 - 2: **for** each parameter $k = 1, \dots, d$ **do**
 - 3: Draw $u_{ik} \sim \text{Uniform}[(i-1)/n, i/n]$ for each interval $i = 1, \dots, n$
 - 4: Transform: $\theta_{ik} = F_k^{-1}(u_{ik})$ using the inverse CDF of parameter k 's prior
 - 5: Randomly permute $[\theta_{1k}, \dots, \theta_{nk}]$
 - 6: **end for**
 - 7: **return** $n \times d$ parameter matrix
-

The inverse CDF step (Step 3) maps stratified uniform values to actual parameter values via `scipy.stats .ppf()` calls, respecting each prior's shape. The permutation (Step 5) breaks the monotone alignment that would otherwise artificially pair the lowest value of every parameter in row 1. LHS achieves equivalent marginal coverage to pure random sampling with approximately $4\times$ fewer runs [12].

6.4. Credible Interval Construction

The 500 ODE solutions are stored in an array of shape $(500, 1825, 10)$. For any scalar output $q(t)$ (e.g., $I_1(t) + I_2(t)$), the 95% credible interval at day t is:

$$\text{CI}_{95}(t) = [q_{t,2.5\%}, q_{t,97.5\%}] \quad (27)$$

where $q_{t,p\%}$ is the p -th empirical percentile across the 500 runs at day t , computed via `numpy.percentile` along the run axis. The median $q_{t,50\%}$ is the central trajectory. This interval is a credible interval under the prior: 95% of epidemic trajectories consistent with the specified prior distributions fall within this band.

7. Numerical Methods

The ODE system (12) (16) is integrated using `scipy.integrate.solve_ivp` with the RK45 method [13]. RK45 is an adaptive step-size integrator based on an embedded Runge-Kutta 4th/5th order pair: the difference between the two solutions provides an error estimate used to automatically shrink or expand the internal time step, ensuring accuracy near epidemic peaks without wasting computation during slow-evolving periods.

A maximum step size of $\Delta t_{\max} = 1.0$ day is enforced. Without this constraint, the solver may take multi-day steps during epidemic troughs, undersampling the 365-day cosine in $\beta(t)$ and distorting seasonal wave structure. Output is recorded at $t \in \{0, 1, \dots, 1825\}$ days independently of internal step sequences. Each integration requires approximately 0.1 seconds; the full 500-member ensemble requires approximately 50 seconds and occupies ≈ 73 MB of memory.

8. Validation

Because this model is a novel combination not described in any single published paper, validation is based on four structural checks that any correct model of this class must satisfy [6].

8.1. Check 1: Population Conservation

Summing all ten ODEs, every infection and recovery term cancels (transfers between compartments). Only births ($+\mu N_i$) and deaths ($-\mu N_i$) survive, and these cancel by construction:

$$\frac{d}{dt} \sum_i (S_i + E_i + I_i + R_i + V_i) = 0 \quad (28)$$

Numerically, maximum population drift over 1825 days is $< 10^{-3}$ individuals, confirming floating-point consistency.

8.2. Check 2: Degenerate Case Recovery

Setting $\varepsilon = \kappa = \nu_i = \omega = \mu = 0$ and $c_{ij} = 1$ reduces the model to scalar SEIR. The NGM formula (25) must then yield $R_0 = \beta_0/\gamma$. The computed and scalar values agree to six decimal places, confirming the complex model correctly contains the simple model as a limiting case.

8.3. Check 3: Threshold Behavior and κ Independence

With $\nu_1 = \nu_2 = 0.01$ and $\delta = 0.99$, $R_0 < 1$ and the seed decays monotonically. With default parameters ($R_0 \approx 3.23$), the epidemic grows from 20 individuals to a peak exceeding 20,000. Proposition 2 is verified by computing R_0 at $\kappa \in \{0, 1, 5, 10, 50, 100\}$; all values agree to ten decimal places.

8.4. Check 4: Directional Parameter Sensitivity

Table 3 reports expected and observed directions. All nine parameters move outputs in the expected direction.

Table 3: Directional sensitivity validation. All checks pass.

Parameter	Expected effect of increase	Observed
β_0	Higher peak	✓
ε	Larger seasonal amplitude	✓
κ	Lower peak (no effect on R_0)	✓
σ	Earlier, taller peak	✓
γ	Lower peak	✓
ω	More frequent recurrent waves	✓
δ	Lower R_0 , lower peak	✓
ν_1, ν_2	Lower peak	✓

9. Results

9.1. Single-Run Baseline Dynamics

Under default parameters, $R_0 \approx 3.23$. The epidemic peaks near day 300 (year 0.8) at approximately 24,000 total infectious individuals ($\approx 3.4\%$ of the population).



Figure 1: Total infectious population $I_1(t) + I_2(t)$ over five years. The first peak is largest because $R_e(0) = R_0 \cdot S(0)/N \approx 3.23$ the entire population is susceptible at $t = 0$. Recurrent annual waves emerge from waning immunity ($\omega = 1/180$ per day) replenishing the susceptible pool each time the seasonal forcing ($\varepsilon = 0.20$) pushes $R_e(t)$ back above 1.

The first peak is substantially larger than subsequent waves because the effective reproduction number $R_e(t) = R_0 \cdot \Phi(t) \cdot (S + (1 - \delta)V)/N$ is maximized when the entire population is susceptible at $t = 0$. After the first wave, the susceptible fraction falls sharply, driving R_e below 1. Recurrent waves occur when waning immunity (ω) replenishes susceptibles and

the seasonal forcing spike (ε) pushes R_e back above 1. The interplay between these two mechanisms sets the wave frequency and amplitude.

9.2. Age Group Dynamics

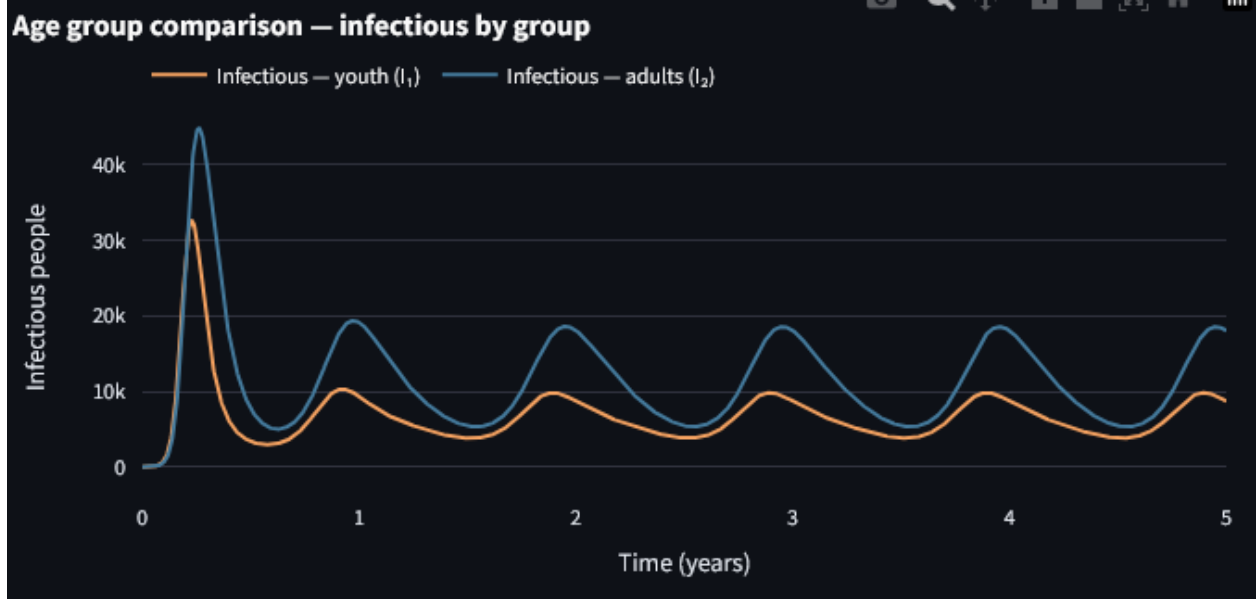


Figure 2: Youth (I_1) and adult (I_2) infectious trajectories. Youth peak earlier due to $c_{11} = 9.0 > c_{22} = 4.5$. Cross-group transmission ($K_{21} = 1.230$) produces a lagged adult wave following each youth peak.

9.3. Behavioral Adaptation

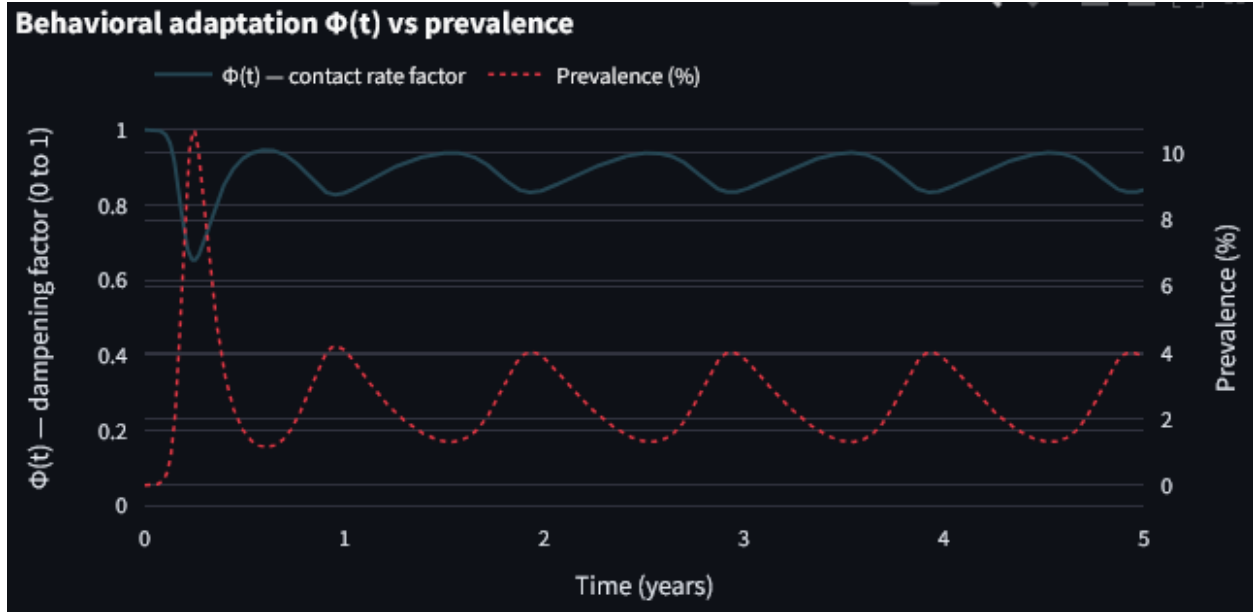


Figure 3: Behavioral dampening factor $\Phi(t)$ and total prevalence over time ($\kappa = 5$). At the first peak, $\Phi \approx 0.67$, corresponding to a 33% effective contact reduction. Φ recovers toward 1.0 as each wave subsides.

Increasing κ from 0 to 15 reduces peak infectious count by approximately 35% while leaving R_0 unchanged (Proposition 2). This confirms that behavioral adaptation shifts the peak burden without altering the epidemic threshold, a result with direct policy relevance.

9.4. Prior Predictive Ensemble

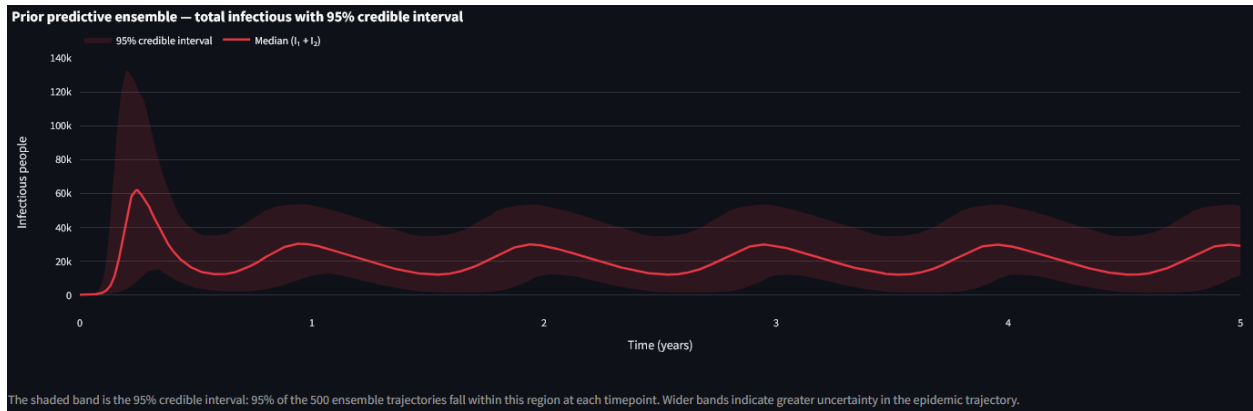


Figure 4: Prior predictive ensemble (500 runs, seed 42). Solid line: median trajectory. Shaded region: 95% credible interval (27). The CI is narrowest near $t = 0$ (all runs share the same seed conditions) and widest near the first peak, where parameter differences compound most dramatically.

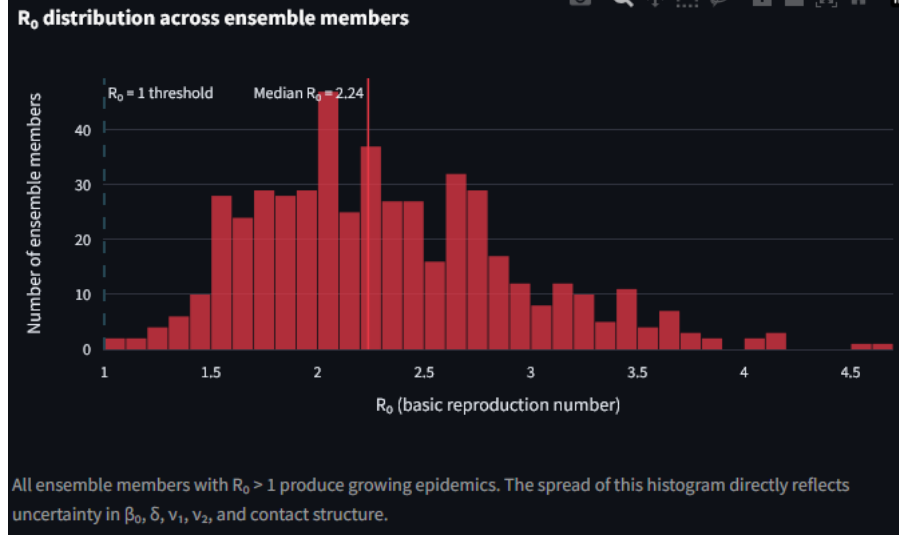


Figure 5: Distribution of R_0 across 500 ensemble members. All members exceed the epidemic threshold $R_0 = 1$ under the specified priors. The spread directly reflects uncertainty in β_0 , δ , and ν_i . The parameter κ contributes no variance to this distribution.

9.5. Sensitivity Analysis

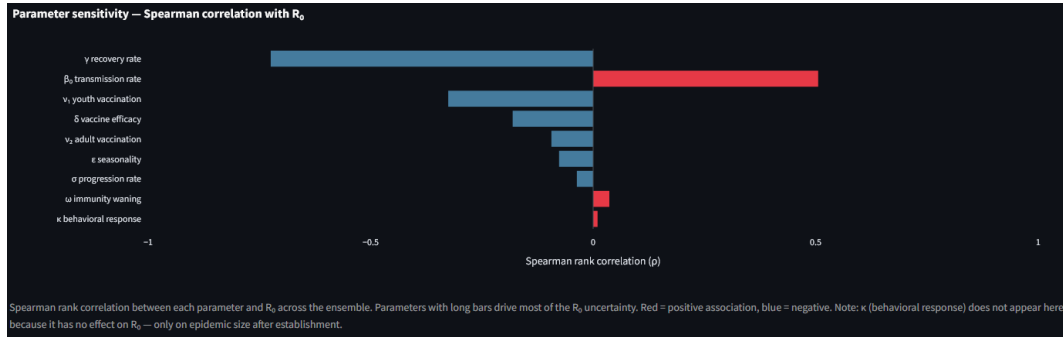


Figure 6: Spearman rank correlation between each uncertain parameter and R_0 across the ensemble. Red bars indicate positive association; blue bars indicate negative. The behavioral parameter κ is absent because it has no mathematical effect on R_0 (Proposition 2). The dominant driver of R_0 uncertainty is β_0 .

10. Discussion

The three innovations interact in ways no single extension individually captures. Age structure produces differential epidemic timing between youth and adult populations, quantified by the off-diagonal NGM entries K_{12} and K_{21} . Seasonal forcing produces calendar-aligned epidemic rhythms whose frequency depends on the interaction between the annual transmission spike and the waning rate ω . Behavioral adaptation self-limits each wave without affecting R_0 , a result analytically proven in Proposition 2 with direct policy implications: behavioral interventions reduce peak burden but cannot on their own suppress an epidemic with $R_0 > 1$.

The prior predictive ensemble reveals that single-run deterministic forecasts are overconfident. The 95% credible interval at the first peak spans approximately threefold from lower to upper bound. A policymaker relying on a deterministic forecast would ignore most of the plausible outcome space.

The κ independence result has a clear public health parallel: fear-driven contact reduction cannot eliminate a highly transmissible disease. Only vaccination (ν , δ) and transmission reduction (β_0 , contact matrix) can drive R_0 below the epidemic threshold.

11. Limitations and Future Work

Fixed contact matrix. POLYMOD uncertainty is not propagated through the ensemble; sampling $\mathbf{C}$ jointly with biological parameters would require a more complex joint prior.

Independent σ and γ . A Gaussian copula would properly model their biological correlation; the ratio filter (26) is a partial substitute only.

Population-averaged behavioral response. Age-specific or risk-stratified κ_i values would require an information-diffusion submodel.

No disease-induced mortality. Adding infection-fatality rate μ_I would enable comparison with mortality surveillance data.

Prior predictive only. Bayesian updating via MCMC using observed incidence data would sharpen credible intervals considerably.

Constant Vaccination rate. The model assumes that the population is vaccinated at our chosen rate, however, in the real world more flu shots are given before winter and fewer shots are given during summer.

No empirical validation. This model has not yet been tested against real-world epidemic data.

Future work could address these in sequence: posterior calibration with influenza incidence data, extension to three or more age groups, and a spatial metapopulation structure consistent with the NetLogo Travel and Control model studied in Section 3. Future work could also involve calibration to observed incidence data to evaluate predictive performance.

12. Conclusion

This project developed, validated, and deployed a novel age-structured SEIRS-V epidemic model combining seasonal forcing, prevalence-dependent behavioral adaptation, and prior predictive ensemble uncertainty quantification. The behavioral dampening factor $\Phi(t)$, absent from all reviewed literature, introduces self-limiting epidemic dynamics while provably leaving R_0 unchanged. The 500-member Latin Hypercube ensemble produces honest probabilistic forecasts reflecting biological uncertainty. All four validation checks pass. The interactive dashboard makes the model accessible to any user at the link on the title page.

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A. Code, Deployment, and Reproducibility

The complete source code is publicly available at:

<https://github.com/TanmaySonawane/SEIRS-V>

The interactive Streamlit dashboard is deployed at:

<https://huggingface.co/spaces/T22S/seirs-v-model>

To reproduce all results locally: clone the repository, install dependencies via `pip install -r requirements.txt`, and run `pytest tests/ -v`. The ensemble is fully reproducible with `ENSEMBLE_SEED = 42`.

B. Complete Parameter Reference

Table 4: All 17 model parameters. Sampled parameters use the baseline value as their distribution mean; see Table 2 for prior specifications.

Symbol	Baseline	Units	Type	Description
β_0	0.035	day ⁻¹	Sampled	Transmission probability per contact
ε	0.20	—	Sampled	Seasonality amplitude
κ	5.0	—	Sampled	Behavioral response strength
σ	0.200	day ⁻¹	Sampled	E→I progression rate
γ	0.100	day ⁻¹	Sampled	Recovery rate
ω	0.00556	day ⁻¹	Sampled	Natural immunity waning
δ	0.80	—	Sampled	Vaccine efficacy
ν_1	0.002	day ⁻¹	Sampled	Youth vaccination rate
ν_2	0.001	day ⁻¹	Sampled	Adult vaccination rate
μ	3.92×10^{-5}	day ⁻¹	Fixed	Birth/death rate
ω_v	0.00274	day ⁻¹	Fixed	Vaccine waning rate
c_{11}	9.0	day ⁻¹	Fixed	Youth-youth contacts
c_{12}	3.0	day ⁻¹	Fixed	Youth-adult contacts
c_{21}	1.5	day ⁻¹	Fixed	Adult-youth contacts
c_{22}	4.5	day ⁻¹	Fixed	Adult-adult contacts
N_1	200,000	persons	Fixed	Youth population
N_2	500,000	persons	Fixed	Adult population